ACS-4901-001

Senior Systems Development Project

IS Director: James Deng
Semester: Fall - Winter 2025-26

Team 3: RNAverse

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1.0 Introduction

RNA methylation is an epigenetic modification critically in regulating biological processes like immunity and cellular development. Different RNA methylations (m6A, m7G, m5C) exist, with m6A being the most common in mammalian mRNA. While ML and DL models can predict methylation status from RNA sequences, many researchers lack coding expertise to train these models. Current tools predict only specific methylations with single ML models.

This project aims to integrate multiple methylations and several trained ML/DL models into a user-friendly web platform allowing researchers to predict RNA methylation sites directly from input RNA sequences, supporting multiple input sequences and downloadable results with visualizations.

1.1 User Environment

- Web-based access via desktop or laptop
- Compatible with major browsers (Chrome, Firefox, Edge)
- Optional lightweight cloud access for compute-intensive ML/DL predictions
- Designed for intuitive navigation, minimal technical setup, and rapid result generation

Users: Researchers studying RNA methylation who have strong biological knowledge but limited coding or machine-learning skills. These users need an easy way to run predictions without writing code.

Environment: Our web based platform provides one simple, browser-assessible environment, running preliminary on the university's ACS server.

Limits / Restrictions: Users may have limited compute resources and limited time, so the system should be lightweight, responsive, and straightforward to use. Results should be downloadable so users can save and manage data on their own systems.

2.0 Current System

2.1 Description

Currently, there is no system present within the University of Winnipeg to predict RNA methylation through a web platform. While there are external tools like m5CStack or other RNA methylation predictors, they are limited to single input of RNA sequence. Other tools which require programming knowledge exist but are not suitable for users without a programming background.

2.2 Function of Current system

- Uses predefined models trained on specific datasets
- Generates prediction outputs with probabilities for each sequence.
- Allows users to download prediction results as a file
- Allows users to specify a Project ID to track their submission.

2.3 Shortcomings Of current Systems

- Existing tools like mCStack(external application) focus only on one methylation type.
- No support for comparing multiple ML/DL models (SVM, CNN, Random Forest, etc.).
- Missing visualization and batch input support
- Do not provide flexibility to expand with new methylation types or models.

3.0 Proposed System

3.1 Description

The proposed system will allow researchers to input one or multiple RNA sequences through a web based platform. It will then process these sequences using the pre-trained machine learning and deep learning models such as SVM, Random Forest, Logistic Regression, MLP, XGBoost, and CNN to predict RNA methylation status. The system will generate these results in both tabular and graphical format which users can download as CSV Files. The interface will be simple and easy to use for non technical users. The new system will be modular which means it will allow future integration of new RNA methylation types.

3.2 Objectives

Developing a web based platform that can accept a single or multiple RNA sequence as input from the user and then predicts methylation status using the pre-trained machine learning and deep learning model such as SVM, Random Forest, Logistic Regression, MLP, XGBoost, and CNN.

- Access to downloadable results in the form of CSV files with visualizations to aid interpretation.
- Making sure that the platform remains modular and maintainable using open-source technologies.
- Provide basic help, guidance, or documentation to support non-technical users.
- Support error handling and input validation to prevent invalid sequence submission.

3.3 Benefits

- **Time-saving**: Researchers can quickly analyze multiple RNA sequences without programming knowledge.
- **Automation**: Eliminates manual preprocessing and external scripting.
- **Reduced Errors**: Standardized prediction workflow reduces computational and human errors.
- **Accessibility**: Expands access to advanced ML/DL predictions to users without coding expertise.
- **Future-proof:** Modular design allows easy addition of new RNA methylation types and updated ML/DL models.

3.4 Scope

In Scope

Feature	**Functionality**
Multi-sequence Input	Accept single or multiple RNA sequences of varying lengths
Integrated ML/DL Models	Predict RNA methylation using pre-trained SVM, RF, LR, MLP, XGBoost, CNN models.
Results Export	Download prediction results as CSV and view graphical visualizations.
Modular Design	Allow future expansion to new methylation types and ML/DL models.
Hosting	Platform deployed on University ACS servers.
Help & Documentation	Provide examples and guidance for non-technical users.

Out of Scope

- User registration or personalized accounts
- Custom reporting feature, which means that users could design their own metric for evaluation is out of scope for this project.
- Predicting RNA methylation beyond the currently available models.
- Mobile-specific layouts/mobile app version of this application is out of scope..

3.5 Methodology

Our project will adopt a hybrid methodology that blends the strengths of both Waterfall and Agile Lifecycles. This approach ensures a solid upfront structure through careful planning and documentation, while also providing flexibility for iterative development, continuous feedback, and adaptation to evolving requirements.

1) Waterfall:
    - Upfront Design and Planning: Clear Architecture, deciding on Tech stack, and choosing datasets and models before beginning to code.
    - Structured Documentation: Requirements, systems flow diagrams, and design documents are established earlier to act as a firm foundation.
    - Fixed Milestones: Key deliverables follow sequential timeline to ensure clarity and accountability.
2) Agile Lifecycle:
    - Iterative Development: The system will be developed in sprints (2-3 weeks each) with incremental features.
    - Continuous Feedback: Sponsor feedback will be taken at the end of each sprint to clarify features and adjust priorities.
    - Modularity & Flexibility: Additional methylation or new functionalities can be easily in each sprints.

3.6 Project Schedule

1. **Project Schedule**

Task Milestone	Deadline
Submission of Team Roles	Week of Sept 2, 2025
Sponsor meeting	Week of Sept 8, 2025
Project Proposal	Week of Sept 26 2025
Plan Review	Week of Nov 10, 2025
Mid-project Review	Week of Jan 12, 2026
Development Review	Week of February 09, 2026
Final Turnover and sign off	Week of March 16, 2026
Seminar and Demo	March 27, 2026
Completion checklist	Week of April 06, 2025

2. **Sprint Planning during development phase**

Sprint	**Purposed Dates**	**Tasks**
Sprint 1-2	Nov10 - Dec 2025	Create data pipeline and input validation (multiple RNA sequences support, error handling, formatting).
Sprint 3-4	Dec 12, 2025 - Jan 23, 2026	Host machine learning models (Support Vector Machine, Random Forrest, Linear Regression) in the backend, and connect them with the user interface to make predictions.
Sprint 5-6	Jan 23 - Feb 27, 2026	Develop visualization features (graphs, plots, and metrics) and downloadable CSV output.
Sprint 7	Feb 27 - Mar 16, 2026	Conduct system integration testing, usability testing, and deployment readiness with final documentation

5.0 Stakeholders & Team members

Stakeholders:

Stakeholders	**Names**
Project Sponsors	Dr. Qian Liu and Dr. Sheela Ramanna
IS Director	Mr. James Deng
Project Team Members	Priyanshu Mittal, Samarjeet Singh, Aniket Thakur and Saif Alam
System/Network Administrator	Nischal Ghorasaini and Drey Rannard
Research Support	Dr. Muhammad Tahir

Team member and their roles:

Name	**Waterfall Role**	**Agile Role**
Priyanshu Mittal	Project Leader	Scrum Master
Samarjeet Singh	Technical Leader & Lead Programmer	Product Owner
Aniket Thakur	Quality Control Specialist & Lead Programmer	Developer
Saif Alam	Lead System Analyst/ Designer	Developer

6.0 Assumptions & Noteworthy Issues

Assumptions:

- The full team of 4 students will remain available throughout the project timeline
- Sponsors will review and approve milestones on time
- IS Director will remain available for consultation and project sign-off
- All tools, libraries, and frameworks will remain open-source and compatible.
- University servers will remain accessible for hosting and deployment.

Noteworthy:

- The platform is currently designed for small to medium datasets; larger inputs may require future enhancements.
- The ML/DL models are pre-trained and may need periodic updates as new RNA methylation data becomes available.

7.0 Conclusion

This project provides a web-based platform for RNA methylation prediction that is simple and accessible for researchers without coding skills. By integrating pre-trained ML/DL models with a user-friendly interface, it enhances efficiency, reduces errors, and makes advanced analysis widely available. In doing so, the platform bridges the gap between biology and computation, opening new opportunities for faster and more reliable research.

8.0 Project Proposal Sign-Off

Role/ Title	Name	Signature	Date
Project Sponsor 1	Dr. Sheela Ramanna		**1 Oct. 2025**
Project Sponsor 2	Dr. Qian Liu		Oct.1, 2025
IS Director	James Deng	JAMES DENG	Oct.1,2025
Team Member 1	Samarjeet Singh	Samarjeet Singh	Oct 2, 2025
Team Member 2	Priyanshu Mittal	P. Mittal	Oct 02, 2025
Team Member 3	Aniket Thakur	Aniket Thakur	Oct 2,2025
Team Member 4	Saif Alam		Oct 02, 2025

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